



All about thinking analytically

Now that you know the five essential skills of a data analyst, you're ready to learn more about what it means to think analytically.

People don't often think about thinking.

Thinking is second nature to us.

It just happens automatically, but there are actually many different ways to think.

Some people think creatively, some think critically, and some people think in abstract ways.

Let's talk about analytical thinking.

Analytical thinking involves identifying and defining a problem and then solving it by using data in an organized, step-by-step manner.

As data analysts, how do we think analytically?

Well, to answer that question, we will now talk about a second set of five.

The five key aspects to analytical thinking.

They are visualization, strategy, problem-orientation,

correlation, and finally,
big-picture and detail-oriented thinking.

Let's start with visualization.

In data analytics, visualization
is the graphical representation of information.

Some examples include graphs,
maps, or other design elements.

Visualization is important because visuals can help
data analysts understand and
explain information more effectively.

Think about it like this.

If you are trying to explain the Grand Canyon to someone,
using words would be much
more challenging than showing them a picture.

A visualization of the Grand Canyon would
help you make your point much quicker.

Now let's talk about the second part of
analytical thinking, being strategic.

With so much data available,
having a strategic mindset is
key to staying focused and on track.

Strategizing helps data analysts see what they
want to achieve with the data and how they can get there.

Strategy also helps improve
the quality and usefulness of the data we collect.

By strategizing, we know all our data is

valuable and can help us accomplish our goals.

Next step on the analytical thinking checklist:

being **problem-oriented**.

Data analysts use a **problem-oriented approach in order to identify, describe, and solve problems.**

It's all about keeping the problem top of mind throughout the entire project.

For example, say a data analyst is told about the problem of a warehouse constantly running out of supplies.

They would move forward with different strategies and processes.

But the number one goal would always be solving the problem of keeping inventory on the shelves.

Data analysts also ask a lot of questions.

This helps improve communication and saves time while working on a solution.

An example of that would be surveying customers about their experiences using a product and building insights from those questions to improve their product.

This leads us to the fourth quality of analytical thinking:

being able to identify

a **correlation** between two or more pieces of data.

A correlation is like a relationship.

You can find all kinds of correlations in data.

Maybe it's the relationship between the length of your hair and the amount of shampoo you need.

Or maybe you notice a correlation between a rainier season leading to

a high number of umbrellas being sold.

But as you start identifying correlations in data, there's one thing you always want to keep in mind:

Correlation does not equal causation.

In other words, just because two pieces of data are both trending in the same direction, that doesn't necessarily mean they are all related.

We'll learn more about that later.

Now the final piece of the analytical thinking puzzle: big-picture thinking.

This means being able to see

the big picture as well as the details.

A jigsaw puzzle is a great way to think about this.

Big-picture thinking is like

looking at a complete puzzle.

You can enjoy the whole picture without getting stuck on every tiny piece that went into making it.

If you only focus on individual pieces, you wouldn't be able to see past that,

which is why big-picture thinking is so important.

It helps you zoom out and

see possibilities and opportunities.

This leads to exciting new ideas or innovations.

On the flip side, detail-oriented thinking is all about

figuring out all of the aspects

that will help you execute a plan.

In other words, the pieces that make up your puzzle.

There are all kinds of problems in

the business world that can benefit from

employees who have both a big-picture

and a detail-oriented way of thinking.

Most of us are naturally better at one or the other.

But you can always develop

the skills to fit both pieces together.

Now that you know the five aspects

of analytical thinking,

visualization, strategy, problem-orientation,

correlation, and

big-picture and detail-oriented thinking,

you can put them to work for

you when you're working with data.

As you continue through this course, you'll learn how.

To execute a plan using detail-oriented thinking, what does a data analyst consider?

- ☐ The big picture
- ☐ The root cause
- ☐ The main idea
- ☒ The specifics

✔ Correct

To execute a plan using detail-oriented thinking, a data analyst considers the specifics.

Explain this topic in simple terms

Let's break down the concept of "Correlation does not equal causation" from the lecture!

This phrase emphasizes that just because two things happen together doesn't mean one causes the other. Imagine noticing that ice cream sales go up on sunny days. Does that mean eating ice cream *causes* sunny weather? Of course not! The sun causes both warmer weather *and* a craving for ice cream. They're correlated (happen together) but one doesn't cause the other.

Give me a summary

his lecture explains **analytical thinking** for data analysts. It covers:

- **Five essential skills:** Visualization, Strategy, Problem-orientation, Correlation, and Big-picture and Detail-oriented thinking.
- **Visualization:** Representing data graphically (like using graphs) for better understanding.
- **Strategy:** Having a plan to achieve goals with data.
- **Problem-orientation:** Focusing on identifying and solving problems using data.
- **Correlation:** Understanding relationships between data points (but remembering that correlation doesn't mean causation!).
- **Big-picture and Detail-oriented thinking:** Being able to see both the overall picture and the smaller details within the data.

Give me real-life examples

Okay, let's dive into a real-life example of how "Correlation does not equal causation" plays out, using the concept of **visualization**:

- **Imagine you're analyzing data for a coffee shop and create a graph showing the number of iced coffee sales and the number of sunglasses sold each month.** You notice a strong correlation - both are higher in the summer months!
- **Does this mean buying sunglasses *causes* people to **crave** iced coffee?** Nope! The real cause is likely the hot weather influencing both purchases.

This example shows why data analysts need to think critically about correlations:

1. **Visualizing the data** helps you spot the relationship between iced coffee and sunglasses sales.
2. **Understanding that "correlation does not equal causation"** prevents you from making wrong assumptions. You wouldn't advise the shop to give away free sunglasses to boost coffee sales!

Give me practice questions

1. **True/False:** A data analyst who is only focused on tiny details of a dataset is practicing good "big-picture thinking."
2. **Multiple Choice:** Which of these is NOT a key aspect of analytical thinking for data analysts? a) Visualization b) Random guessing c) Strategy d) Problem-orientation
3. **Open-Ended:** Why is it important for data analysts to use visualization techniques when working with data?
4. **Open-Ended:** The lecture mentions that "correlation does not equal causation." Briefly explain what this means in your own words.
5. **True/False:** If a data analyst discovers a correlation, it means they've definitively proven that one data point causes the other to change.